

Guillaume Dalle

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Researcher in applied mathematics



Professional Experience

Researcher

ÉCOLE NATIONALE DES PONTS ET CHAUSSÉES

2025 - present

Champs-sur-Marne, France

- Applied mathematics for transportation systems

Postdoctoral researcher

ÉCOLE POLYTECHNIQUE FÉDÉRALE DE LAUSANNE

2023 - 2024

Lausanne, Switzerland

- Theoretical study of graph neural networks

Research intern

ÉLECTRICITÉ DE FRANCE R&D

2018

Chatou, France

- Markovian models for soiling estimation & cleaning optimization in solar power plants

Software development intern

FORIS AI

2017

Santiago, Chile

- Linear Programming & local search for university timetabling

Education

PhD in applied mathematics

ÉCOLE NATIONALE DES PONTS ET CHAUSSÉES

2019 - 2022

Champs-sur-Marne, France

- Thesis: *Machine Learning and Combinatorial Optimization Algorithms, with Applications to Railway Planning*

Master of Science in applied mathematics

ÉCOLE NORMALE SUPÉRIEURE PARIS-SACLAY

2018 - 2019

Cachan, France

- Coursework in machine learning and optimization
- Thesis: *Delay propagation on suburban railway networks*

Engineering degree

ÉCOLE POLYTECHNIQUE

2015 - 2018

Palaiseau, France

- Coursework in applied mathematics, computer science and humanities

Scientific preparatory classes

LYCÉE LOUIS-LE-GRAND

2013 - 2015

Paris, France

- Coursework in mathematics and physics

Bilingual French-German high school

LYCÉE JEANNE D'ARC

2010 - 2013

Clermont-Ferrand, France

Funded projects

ACME (Approvisionnement Collaboratif Multimodal Écologique)

PEPR MOBIDEC - AGENCE NATIONALE DE LA RECHERCHE

2025 - 2031

- Multi-disciplinary research consortium to foster mutualization in the supply chain.
- Project website: <https://acme-mobidec.github.io/>

Publications

- [1] L. Bouvier, G. Dalle, A. Parmentier, and T. Vidal, "Solving a Continent-Scale Inventory Routing Problem at Renault," *Transportation Science*, Oct. 2023, doi: [10.1287/trsc.2022.0342](https://doi.org/10.1287/trsc.2022.0342).
- [2] L. Clarté, A. Vandenbroucq, G. Dalle, B. Loureiro, F. Krzakala, and L. Zdeborova, "Analysis of Bootstrap and Subsampling in High-dimensional Regularized Regression," in *The 40th Conference on Uncertainty in Artificial Intelligence*, Jun. 2024. [Online]. Available: <https://openreview.net/forum?id=yZaXk3OxVS>
- [3] G. Dalle and A. Hill, "A Common Interface for Automatic Differentiation," *Journal of Machine Learning Research*, vol. 27, no. 25, pp. 1–13, 2026, [Online]. Available: <http://jmlr.org/papers/v27/25-1024.html>
- [4] G. Dalle, "HiddenMarkovModels.Jl: Generic, Fast and Reliable State Space Modeling," *Journal of Open Source Software*, vol. 9, no. 96, p. 6436, Apr. 2024, doi: [10.21105/joss.06436](https://doi.org/10.21105/joss.06436).

- [5] G. Dalle, L. Baty, L. Bouvier, and A. Parmentier, “Learning with Combinatorial Optimization Layers: A Probabilistic Approach.” [Online]. Available: <http://arxiv.org/abs/2207.13513>
- [6] G. Dalle, “Machine Learning and Combinatorial Optimization Algorithms, with Applications to Railway Planning,” Doctoral dissertation, 2022. [Online]. Available: <https://pastel.hal.science/tel-04053322>
- [7] G. Dalle and Y. D. Castro, “Minimax Estimation of Partially-Observed Vector Autoregressions,” *Electronic Journal of Statistics*, vol. 19, no. 1, pp. 2364–2410, Jan. 2025, doi: [10.1214/25-EJS2387](https://doi.org/10.1214/25-EJS2387).
- [8] G. Dalle and P. Thiran, “Optimal Performance of Graph Convolutional Networks on the Contextual Stochastic Block Model,” in *Proceedings of the Third Learning on Graphs Conference*, PMLR, Jul. 2025, p. 21:1–21:17. [Online]. Available: <https://proceedings.mlr.press/v269/dalle25a.html>
- [9] A. Hill, G. Dalle, and A. Montoison, “An Illustrated Guide to Automatic Sparse Differentiation,” in *ICLR Blogposts 2025*, Apr. 2025. [Online]. Available: <https://iclr-blogposts.github.io/2025/blog/sparse-autodiff/>
- [10] A. Hill and G. Dalle, “Sparser, Better, Faster, Stronger: Sparsity Detection for Efficient Automatic Differentiation,” *Transactions on Machine Learning Research*, 2025, [Online]. Available: <https://openreview.net/forum?id=GtXSN52nIW>
- [11] A. Montoison, G. Dalle, and A. Gebremedhin, “Revisiting Sparse Matrix Coloring and Bicoloring.” [Online]. Available: <http://arxiv.org/abs/2505.07308>
- [12] P. Stephan and G. Dalle, “Method for Determining a Soiling Speed of a Photovoltaic Generation Unit,” WO2020115431A1, Jun. 2020 [Online]. Available: <https://patents.google.com/patent/WO2020115431A1/en>

Talks

Conferences

2025

- RT Optimisation: *Linear and combinatorial optimization on GPUs* (poster session)
- Royal Statistical Society International Conference: *Teaching with Julia* (abstract)
- ROADEF: *Decision-focused learning: software meets theory* (abstract)

2024

- PyData Global: *Automatic differentiation: a tale of two languages* (abstract)
- Julia Optimization Days: *Automatic differentiation: Julia's most confusing superpower?* (abstract)
- JuliaCon: *Gradients for everyone: a quick guide to autodiff in Julia* (abstract)
- JuliaCon: *Fast and generic Hidden Markov Models* (abstract)
- JuliaCon: *Interfaces.jl: base and package interface tests for your objects* (abstract)

2023

- Workshop “Exploring synergies: Machine Learning Meets Physics & Optimization”: *Every solution, everywhere, all at once: turning optimization solvers into probability distributions* (abstract)
- Journées Polyèdres et Optimisation Combinatoire: *What is the gradient of a Linear Program? Automatic differentiation on a polytope*
- Julia and Optimization Days: *Graphs in Julia - on the edge of glory* (slides)
- Julia and Optimization Days: *Writing fast Julia code* (slides)

2022

- JuliaCon: *InferOpt.jl: combinatorial optimization in ML pipelines* (recording)
- JuliaCon: *ImplicitDifferentiation.jl: differentiating implicit functions* (recording)
- Journées SMAI-MODE: *Recherche d'itinéraires dans un réseau ferroviaire : apprendre à mieux optimiser* (abstract)
- ROADEF: *Learning to Solve Stochastic Multi-Agent Path Finding* (abstract)

2020

- ROADEF: *Delay propagation on a suburban railway network* (abstract)

2019

- PGMO Days: *Delay propagation on a suburban railway network* (abstract)

Invitations & seminars

2025

- Cycle de conférences SMAI & Musée des arts et métiers: *Des mathématiques pour faire rouler des trains* (web page, slides)
- AdONE Summer School on Optimization-augmented Machine Learning: *Introduction to automatic differentiation & Advanced automatic differentiation*
- Laboratoire Jean Kuntzmann, Grenoble: *Sparser, better, faster, stronger - automatic differentiation with a lot of zeros*
- Laboratoire Ville Mobilité Transport: *Mixing data, optimization and decision: the role of automatic differentiation*
- Califrais, Paris: *A tutorial on algorithmic differentiation*
- Julia Meetup Paris: *Automatic differentiation: a tale of two languages*
- Inria Paris: *Sparse automatic differentiation - from theory to practice*

2024

- INDY lab, EPFL: *Sparse automatic differentiation - the fastest Jacobians in the West*
- INDY lab, EPFL: *A primer on algorithmic differentiation*
- Institut de Mathématiques de Bourgogne: *Deep learning meets combinatorial optimization - a tale of missing gradients* (abstract)

2023

- Forum Entreprises & Mathématiques: *Railway to shell: des trains, des maths et du code*
- Swiss Data Science Center: *Every solution, everywhere, all at once: turning optimization solvers into probability distributions*

2022

- MIT CSAIL: *Learning with Combinatorial Optimization Layers: A Probabilistic Approach*

2021

- Institut de Recherche Mathématique de Rennes: *Pourquoi les trains sont-ils toujours en retard?* ([abstract](#))
- École Nationale des Ponts et Chaussées: *Understanding railway delay propagation through latent variable models* ([abstract](#))

Teaching

Teaching assistant & guest lecturer

ÉCOLE NATIONALE DES PONTS ET CHAUSSÉES

2025 - present

Champs-sur-Marne, France

- APOC: Deep learning for combinatorial optimization ([course page](#))
- DECO: Decomposition methods for integer programming ([course page](#))
- MCNDU: Mobilité connectée : nouvelles données, nouveaux usages ([course page](#))
- IOPTI: Introduction to optimization ([course page](#))

Teaching assistant

ÉCOLE POLYTECHNIQUE FÉDÉRALE DE LAUSANNE

2023 - present

Lausanne, Switzerland

- Statistical physics for optimization & learning ([course page](#))
- Modèles stochastiques pour les communications ([course page](#))

Teaching assistant & guest lecturer

MASSACHUSETTS INSTITUTE OF TECHNOLOGY

2022 - 2023

Cambridge (MA), United States

- Julia - solving real-world problems with computation ([course page](#))

Teaching assistant & lecturer

ÉCOLE NATIONALE DES PONTS ET CHAUSSÉES

2019 - 2022

Champs-sur-Marne, France

- Introduction to optimization
- Operations research ([course page](#))
- Design of optimization challenges on:
 - facility location with Air Liquide
 - train shunting with SNCF
 - inventory routing with Renault

Civic service

LYCÉE JEAN-BAPTISTE COROT

2015

Savigny-sur-Orge, France

- Scientific courses and mentoring for struggling high school and undergraduate students

Mentoring

PhD supervisor

ÉCOLE NATIONALE DES PONTS ET CHAUSSÉES

2026 - ongoing

Champs-sur-Marne, France

- [Nicolas Bessin](#) (2026-2029, co-supervised with Frédéric Meunier): *Optimizing mutualization in the food supply chain*
- [Simeon Schaub](#) (2026-2029, co-supervised with Axel Parmentier): *Combinatorial optimization on GPUs for decision-focused learning*

Student advisor

ÉCOLE NATIONALE DES PONTS ET CHAUSSÉES

2025 - ongoing

Champs-sur-Marne, France

- [Yasmin Van Den Broek](#) (2025): *A Decision-Focused Learning Approach for Multi-Agent Pathfinding*
- Various group projects:
 - Optimized dorm room assignment (2022)
 - Pathfinding in historical maps (2022)
 - Multimodal transit routing (2020)
 - Train shunting (2020)

Student advisor

ÉCOLE POLYTECHNIQUE FÉDÉRALE DE LAUSANNE

2023 - 2025

Lausanne, Switzerland

- [Antoine Buttier](#) (2025): *Accelerated traffic assignment*
- [Antoine Buttier](#) (2024): *Accelerating sparse linear algebra and GraphBLAS with GPUs*
- [Robin Bochatay](#) (2024): *Speeding up Railway Generation and Train Simulation for the Flatland Challenge*
- [Oleg Fafurin](#) (2024): *Speeding up Railway Generation and Train Simulation for the Flatland Challenge*
- [Chun-Tso Tsai](#) (2023): *Multi-Agent Pathfinding with Mixed-Integer Linear Programming and Lagrange Relaxation*
- Various group and individual projects:
 - Experimentation with parallel graph algorithms (2024)
 - Flatland challenge (2024)
 - graph community detection (2023)

Student advisor

ÉCOLE NATIONALE DES PONTS ET CHAUSSÉES

2019 - 2022

Champs-sur-Marne, France

- [Louis Bouvier](#) (2021): *Large Neighborhood Search and Structured Prediction for the Inventory Routing Problem*
- Various group projects:
 - Optimized dorm room assignment (2022)
 - Pathfinding in historical maps (2022)
 - Train shunting (2020)
 - Multimodal transit routing (2020)

Community service

Action Editor

SCIENTIFIC PUBLISHING

ongoing

- Action Editor for [Transactions on Machine Learning Research](#)

Reviewer

SCIENTIFIC PUBLISHING

ongoing

- Reviewer for scientific journals: [Transactions on Machine Learning Research](#) (expert reviewer), [Journal of Open Source Software](#), [Transportation Science](#), [Transportation Research](#)
- Reviewer for conferences: [NeurIPS](#) (2023, 2025), [ICML](#) (2023), [Learning on Graphs](#) (2024, 2025), [JuliaCon](#) (2021-2026)

PhD jury member

VARIOUS UNIVERSITIES

2025-ongoing

- [Aurora Rossi](#) (2025): Computational Methods and Analysis of Temporal Networks
- [Léo Baty](#) (2025): Combinatorial optimization and decision-focused learning: algorithms, implementations, and applications at Air France

Conference organizer

JULIACON LOCAL

October 2025

Paris

Chair of [JuliaCon Local Paris 2025](#), a european conference on scientific computing and open-source software (170 participants)

Conference organizer

ROADEF

February 2025

Champs-sur-Marne

Co-chair of [ROADEF 2025](#), the national French conference on operations research (>500 participants)

PhD representative

ÉCOLE DES PONTS

2021

Champs-sur-Marne

Representative for PhD students at my university. Advocated for student well-being during Covid lockdowns.

Awards

- | | |
|------|---|
| 2025 | Open science prize for research software , Ministry for higher education and research |
| 2024 | Julia Community Prize , JuliaCon |
| 2024 | EURO Excellence in Practice Award (finalist) , The Association of European Operations Research Societies |
| 2023 | PhD award for mathematics in industry , AMIES |
| 2021 | Prix Pasquet for an outstanding engineering student , École des Ponts ParisTech |
| 2015 | Ranked first in the nationwide entrance exam , École polytechnique |

Skills

Languages	English French German Spanish
Programming	Julia Python Rust Git(Hub) LaTeX Typst
Personal Interests	Music Songwriting Board games Running Bouldering